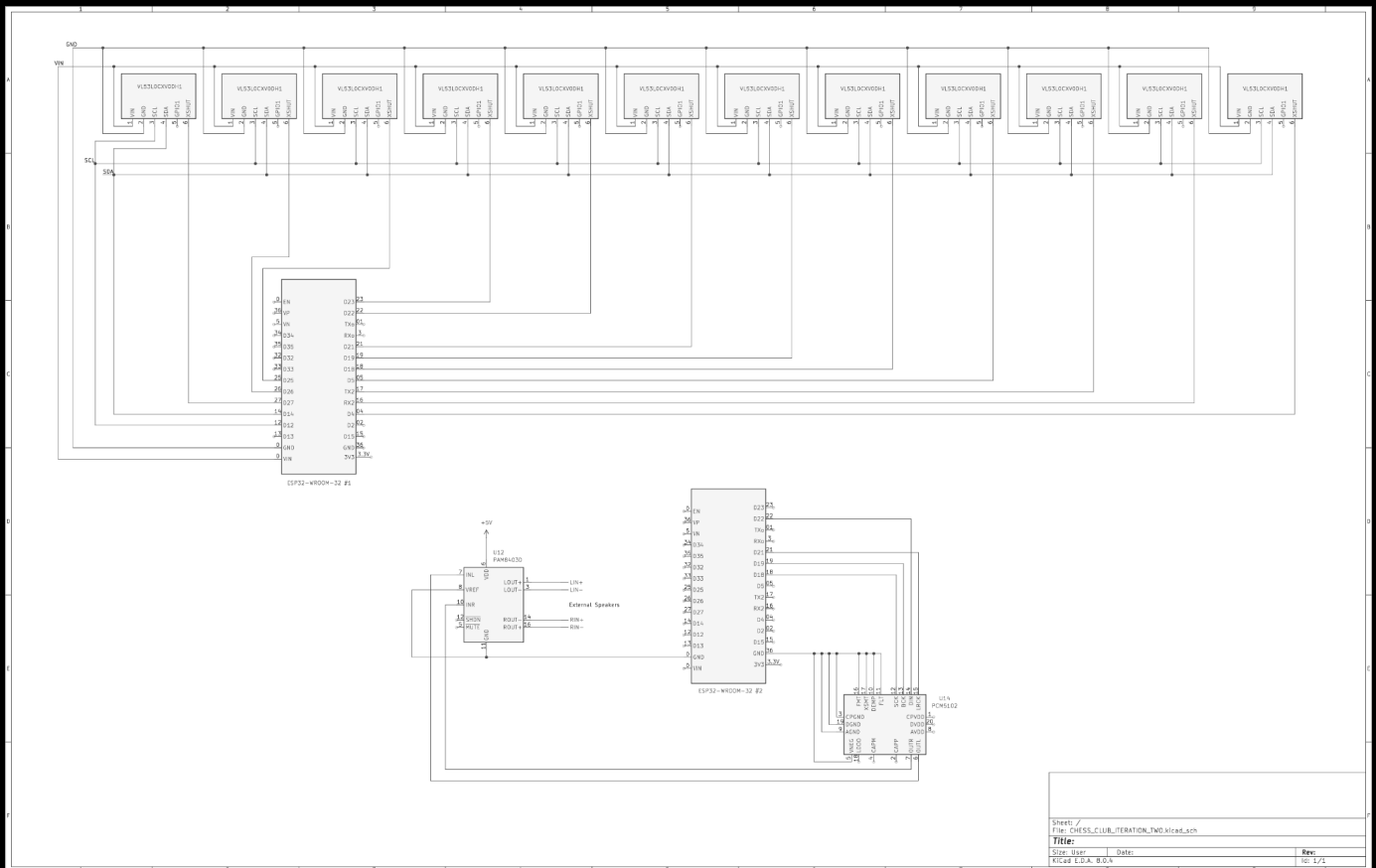


develop flow from OUTPUT to INPUT

Complete prototype testing in this order:

1. Speaker output
2. .wav to speaker
3. .wav to ESP32 filter to speaker
4. ESP32 waveform generation to speaker
5. ESP32 waveform generation to ESP32 filter to speaker



> Schematic overview

* power

- 3.3V power supply
- VCC + GND

* inputs

- sensors
 - VIN to 3.3V
 - GND to GND
 - SCL to ESP32 SCL
 - SDA to ESP32 SDA
 - GPIO (none)
 - XSHUT to unique GPIO on ESP32
- microphone

* processors

- ESP32s
- amplifiers

* outputs

- speakers
- subwoofer
- oled display

* Component Specifications

(7) ESP32-ROVER-E

[documentation](https://www.espressif.com/sites/default/files/documentation/esp32-wrover-e_esp32-wrover-ie_datasheet_en.pdf)

- 2 data processors, 3 audio processors, 1 oscilloscope processor, 1 audio output/master coordinator.
- 3.3V
- 8MB PSRAM, 8MB Flash
- Wifi & Bluetooth

(11) VL53L4CX Time-of-Flight Sensor [listing](<https://www.adafruit.com/product/5425>)

- 3.3V
- I2C communication

- SCL, SDA, GPIO, XSHUT pins

(1) MAX9814 microphone

[documentation](<https://cdn-shop.adafruit.com/datasheets/MAX9814.pdf>)

(2) MAX98357 Amplifier

[documentation](<https://learn.adafruit.com/adafruit-max98357-i2s-class-d-mono-amp>)

(1) 128x64 STEMMMA QT OLED display

[documentation](<https://learn.adafruit.com/monochrome-oled-breakouts>)

>**Power & GND**

- 20 power (VDD) terminals to 3.3V
- 7 ESP32s, 11 sensors, 1 microphone, 1 OLED display
- External power for subwoofer & speakers

>**Reference**

microphone and audio processing

[library](<https://github.com/sheavey/ESP32-AudioInI2S>)

> Allocation

ESP32 Processor #1: Data collection from ToF sensors

Processor #2: Data collection (...)

#3: Waveform synthesizer & basic manipulations

#4: Undecided waveform manipulation

#5: Undecided waveform manipulation

#6: Oscillator generation for OLED display

#7 Audio output processor & mixer for speakers

<https://github.com/hamuro80/blackstomp>

A quick development platform for ESP32-A1S digital effect pedal...!

- *Taptempo Stereo (Ping-Pong) Delay*
- *Stereo Chorus*
- *BLE Pedal*
- *MIDI Pedal*
- *Mic Mixer*
- *Gain Doubler*
- *Distortion*